

Climate Control System



WARNING: Air conditioning liquid refrigerant R-134a and R-1234yf are capable of harming eyes or freezing skin. Always wear safety goggles and avoid contact with liquid refrigerant. Failure to follow these instructions may result in serious personal injury.

WARNING: Air conditioning refrigerant R-134a and R-1234yf gases are capable of displacing the air necessary for breathing in a confined space and can produce toxic gasses if burned at high temperatures. Do not release a large quantity of R-134a or R-1234yf gas within a confined space or allow it to be exposed to a high temperature flame (such as a torch). Consult the Material Safety Data Sheet (MSDS) for further information. Failure to follow these instructions may result in serious personal injury or death.

NOTICE: To avoid damaging the vehicle or A/C components, the following precautions must be observed.

- The A/C refrigerant of all vehicles must be identified and analyzed prior to refrigerant charging. Failure to do so can contaminate the shop bulk refrigerant and other vehicles.
- Do not add R-12 refrigerant to an A/C system that requires the use of R-134a refrigerant. These 2 types of refrigerant must never be mixed. Doing so can damage the A/C system.
- Charge the A/C system with R-134a refrigerant gas while the engine is running only at the low-pressure side to prevent refrigerant slugging from damaging the A/C compressor.
- Use only R-134a refrigerant. Due to environmental concerns, when the A/C system is drained, the refrigerant must be collected using refrigerant recovery/recycling equipment. Federal, State/Provincial and/or local laws REQUIRE that R-134a be recovered into appropriate recovery equipment and the process be conducted by qualified technicians who have been certified by an approved organization, such as ASE or MACS . Use of a recovery machine dedicated to R-134a is necessary to reduce the possibility of oil and refrigerant incompatibility concerns. Refer to the instructions provided by the equipment manufacturer when removing refrigerant from or charging the A/C system.
- Refrigerant R-134a must not be mixed with air for leak testing or used with air for any other purpose above atmospheric pressure. R-134a is combustible when mixed with high concentrations of air and higher pressures.
- A number of manufacturers are producing refrigerant products that are described as direct substitutes for refrigerant R-134a. The use of any unauthorized substitute refrigerant can severely damage the A/C components. If repair is required, use only new or recycled refrigerant R-134a.

NOTICE: To avoid contamination of the A/C system, the following precautions must be observed.

- Never open or loosen a connection before recovering the refrigerant.
- When loosening a connection, if any residual pressure is evident, allow it to leak out before opening the fitting.
- Evacuate a system that has been opened to install a new component or one that has discharged through leakage before charging.
- Seal open fittings with a cap or plug immediately after disconnecting a component from the system.
- Clean the outside of the fittings thoroughly before disconnecting a component from the system.
- Do not remove the sealing caps from a new component until ready to install.
- Refrigerant oil absorbs moisture from the atmosphere if left uncapped. Do not open an oil container until ready to use and install the cap immediately after using. Store the oil in a clean, moisture-free container.
- Install a new O-ring seal before connecting an open fitting. Coat the fitting and O-ring seal with PAG oil before connecting.
- When installing a refrigerant line, avoid sharp bends. Position the line away from the exhaust or any sharp edges that can chafe the line.
- Tighten threaded fittings only to specifications. The steel and aluminum fittings used in the refrigerant system does not tolerate overtightening.

- When disconnecting a fitting, use a wrench on both halves of the fitting to prevent twisting of the refrigerant lines or tubes.
- Do not open a refrigerant system or uncap a new component unless it is as close as possible to room temperature. This prevents condensation from forming inside a component that is cooler than the surrounding air.

The system consists of the following components:

- A/C compressor
- A/C compressor clutch assembly
- A/C condenser core
- A/C evaporator core
- Receiver/drier
- Connecting refrigerant lines
- TXV
- A/C pressure transducer
- Heater core and evaporator core housing
- HVAC module
- FCIM
- FCDIM (non touchscreen only)
- FDIM (touchscreen only)
- Air inlet mode door actuator
- Defrost/panel/floor mode door actuator
- Temperature blend door actuator(s)
- Blower motor resistor (EMTC base systems only)
- Blower motor speed control (remote HVAC control module systems only)
- Blower motor
- In-vehicle temperature and humidity sensor (remote HVAC control module systems only)
- Sunload sensor (EATC dual zone systems only)
- Evaporator temperature sensor (serviced only with the heater core and evaporator core housing)

External Temperature Display

The ambient air temperature sensor is monitored by the PCM. The PCM sends the raw value through the HS-CAN to the BCM. The BCM sends the data through the MS-CAN to the HVAC module. The HVAC module filters and stabilizes the value and sends it back to the BCM. The BCM sends the filtered ambient temperature data to the IPC over the I-CAN. The IPC sends the data over the I-CAN to the FDIM to display the temperature.

While the actual external temperature can vary continuously, the value that is displayed updates at a specific rate depending on whether or not the engine is "hot" and whether or not the vehicle is moving.

When the external temperature is rising, the display updates slowly.

Updates (when the actual external temperature is higher than currently displayed value) is limited to 0.6°C (1°F) every 20 seconds while the vehicle is moving at greater than 32 km/h (20 mph) for more than 60 seconds or 0.6°C (1°F) every 20 minutes if the vehicle is not moving at this speed. This is to prevent the heat from the engine compartment from affecting the accuracy of the display.

When the external temperature is dropping, the display updates quickly.

Updates (when the actual external temperature is lower than currently displayed value) are limited to 0.6°C (1°F) every 2 seconds (regardless of vehicle speed). Consequently the display essentially follows the drop experienced by the external temperature thermistor.

EMTC — Base System

System Operation

The EMTC base system heats or cools the vehicle depending on the HVAC module control panel selection. The EMTC base system has 4 blower motor speed positions.

- The control panel selection determines heating or cooling, air distribution and enables blower motor operation.
- The temperature control setting determines the outlet air temperature.
- The blower motor switch varies the blower motor speed (4 speed positions).
- During A/C operation, the system reduces the relative humidity of the air inside the vehicle.

Remote Start System Behavior — Message Center Set to Auto

Remote start is an available feature on this vehicle. In addition to being able to start the vehicle remotely, the remote start feature also utilizes other vehicle systems to increase the level of comfort to the vehicle occupants upon entering the vehicle. For additional information on the remote start feature and the other vehicle systems it utilizes, refer to Section 303-06.

When the remote start feature is used, the climate control system automatically sets certain parameters in an attempt to achieve a comfortable cabin temperature. These parameters are set based on outside air temperature. During remote start, the outside air temperature is continually evaluated and HVAC system behavior can change if the outside air changes between cold, moderate and warm temperatures.

For cold ambient air temperatures (below 4.4°C (40°F)):

- the airflow mode is set to FLOOR/DEFROST if Front Defrost is set to Off in the message center.
- the airflow mode is set to DEFROST if Front Defrost is set to Auto in the message center.
- the temperature is set to full HOT.
- the blower speed is set to HIGH.
- the air inlet mode is set to fresh air.
- A/C is set to OFF.

For moderate ambient air temperatures (between 4.4°C (40°F) and 23.9°C (75°F)):

- the airflow mode is set to the last user setting.
- the temperature is set to the last user setting.
- the blower speed is set to the last user setting.
- the air inlet mode is set to the last user setting.
- A/C is set to the last user setting.

For warm ambient air temperatures (above 23.9°C (75°F)):

- the airflow mode is set to MAX A/C.
- the temperature is set to full COOL.
- the blower speed is set to HIGH.
- the air inlet mode is set to RECIRC.
- A/C is set to ON.

Remote Start System Behavior — Message Center Set to Last User Settings

Remote start is an available feature on this vehicle. In addition to being able to start the vehicle remotely, the remote start feature also utilizes other vehicle systems to increase the level of comfort to the vehicle occupants upon entering the vehicle. For additional information on the remote start feature and the other vehicle systems it utilizes, refer to Section 303-06.

When the remote start feature is used, the climate control system automatically sets certain parameters in an attempt to achieve a comfortable cabin temperature. These parameters are set based on outside air temperature. During remote start, the outside air temperature is continually evaluated and HVAC system behavior can change if the outside air changes between cold, moderate and warm temperatures.

For additional information regarding the message center, refer to Section 413-01.

When Heater — A/C is set to Last User Settings in the message center, the remote start sets the climate control system to the following settings.

- For cold ambient air temperatures as determined by the HVAC module:
 - the airflow mode is set to the last user setting if Front Defrost is set to Off in the message center.
 - the airflow mode is set to DEFROST if Front Defrost is set to Auto in the message center.

- the temperature is set to the last user setting.
- the blower speed is set to the last user setting.
- the air inlet mode is set to the last user setting.
- A/C is set to the last user setting.
- For moderate and warm ambient air temperatures as determined by the HVAC module:
 - the airflow mode is set to the last user setting.
 - the temperature is set to the last user setting.
 - the blower speed is set to the last user setting.
 - the air inlet mode is set to the last user setting.
 - A/C is set to the last user setting.

System Airflow and Mode Description

Max A/C

When MAX A/C is selected:

- the air inlet door closes, preventing outside air and admits only recirculated air.
- the RECIRC indicator is illuminated (RECIRC forced on).
- the mode doors direct airflow to the instrument panel A/C registers.
- blended air temperature is available.
- the A/C request button is illuminated.
- the A/C compressor operates if the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

PANEL

When PANEL is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the instrument panel A/C registers.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

PANEL/FLOOR

When PANEL/FLOOR is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the floor duct and the instrument panel A/C registers. A small amount of airflow from the side window demisters and defrost duct is present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

OFF

When OFF is selected:

- the air inlet door closes, preventing outside air and allowing only recirculated air.
- the blower motor is off.

FLOOR

When FLOOR is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment. In this mode, the RECIRC request times-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.
- the mode doors direct airflow to the floor duct. A small amount of airflow from the defroster duct and side window demisters is present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

FLOOR/DEFROST

When FLOOR/DEFROST is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment. In this mode, the RECIRC request times-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.
- the mode doors direct airflow to the floor duct, the defroster duct and the side window demisters.
- blended air temperature is available.
- the A/C request button toggles the indicator on and off. To reduce fogging, the A/C compressor operates automatically, regardless of indicator status, if the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

DEFROST

When DEFROST is selected:

- the RECIRC request button is disabled. The air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the defroster duct and side window demisters. A small amount of airflow from the floor duct is present.
- blended air temperature is available.
- the A/C request button toggles the indicator on and off. To reduce fogging, the A/C compressor operates automatically, regardless of indicator status, if the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

HVAC Module — Remote Mount

The EATC dual zone and the EMTC 4.2 inch display systems use a remote HVAC module that is separate from the control interface. The EMTC with 4.2 inch display system has 7 blower motor speed positions.

System Operation

The system maintains the selected vehicle interior temperature by heating and/or cooling the air.

- During A/C operation, the system reduces the relative humidity of the air inside the vehicle.
- The driver may override the automatic mode of operation.
- The temperature control setting is the desired cabin air temperature.
- The blower motor control override buttons vary the blower motor speed.
- The driver side and passenger side temperature settings can be individually controlled (dual zone systems only).

When the system is set to AUTO the air distribution direction, blower motor speed and A/C compressor operation and air inlet door mode are automatically controlled based on the temperature(s) selected.

The system manual override settings:

- allow the air distribution direction to be manually selected.
- allow the blower motor speed to be manually selected.
- allow recirculation mode to be manually selected in all modes except DEFROST. The A/C request can be manually selected by pressing the A/C request switch.
- allow A/C compressor operation to be manually selected except in DEFROST and FLOOR/DEFROST.

Load Shed Behavior

Under certain circumstances, the vehicle can enter an engine running load shed mode. When this condition occurs, the blower motor in the EATC system behaves in the following manner:

- If the blower motor speed setting is greater than 50%, the speed automatically is reduced to 50% of the maximum setting.
- If the blower motor speed setting is equal to or less than 50%, the speed automatically is reduced to the next lowest level.
- If the blower motor speed setting is set to its lowest setting, it remains in the lowest setting.

For information on when the vehicle can enter an engine running load shed mode, refer to Section 414-00.

Remote Start System Behavior — Message Center Set to Auto

Remote start is an available feature on this vehicle. In addition to being able to start the vehicle remotely, the remote start feature also utilizes other vehicle systems to increase the level of comfort to the vehicle occupants upon entering the vehicle. For additional information on the remote start feature and the other vehicle systems it utilizes, refer to Section 303-06.

When the remote start feature is used, the climate control system automatically sets certain parameters in an attempt to achieve a comfortable cabin temperature. These parameters are set based on outside air temperature. During remote start, the outside air temperature is continually evaluated and HVAC system behavior can change if the outside air changes between cold, moderate and warm temperatures.

For additional information regarding the message center, refer to Section 413-01.

When Heater — A/C is set to Auto in the message center, the remote start sets the climate control system to the following settings:

- For cold ambient air temperatures as determined by the HVAC module AUTO mode to request heat:
 - the airflow mode is set to AUTO if Front Defrost is set to Off in the message center. (if the HVAC module requests FLOOR in AUTO mode, FLOOR/DEFROST is used in remote start mode).
 - the airflow mode is set to DEFROST if Front Defrost is set to Auto in the message center.
 - the temperature is set to 22°C (72°F).
 - the blower speed is set to AUTO.
 - the air inlet mode is set to AUTO.
 - A/C is set to AUTO.
- For moderate and warm ambient air temperatures as determined by the HVAC module AUTO mode to request heat or cooling:
 - the airflow mode is set to AUTO.
 - the temperature is set to 22°C (72°F).
 - the blower speed is set to AUTO.
 - the air inlet mode is set to AUTO.
 - A/C is set to AUTO.

Remote Start System Behavior — Message Center Set to Last User Settings

Remote start is an available feature on this vehicle. In addition to being able to start the vehicle remotely, the remote start feature also utilizes other vehicle systems to increase the level of comfort to the vehicle occupants upon entering the vehicle.

For additional information on the remote start feature and the other vehicle systems it utilizes, refer to Section 303-06.

When the remote start feature is used, the climate control system automatically sets certain parameters in an attempt to achieve a comfortable cabin temperature. These parameters are set based on outside air temperature. During remote start, the outside air temperature is continually evaluated and HVAC system behavior can change if the outside air changes between cold, moderate and warm temperatures.

For additional information regarding the message center, refer to Section 413-01.

When Heater — A/C is set to Last User Settings in the message center, the remote start sets the climate control system to the following settings:

- For cold ambient air temperatures as determined by the HVAC module:
 - the airflow mode is set to the last user setting if Front Defrost is set to Off in the message center.
 - the airflow mode is set to DEFROST if Front Defrost is set to Auto in the message center.
 - the temperature is set to the last user setting.
 - the blower speed is set to the last user setting.
 - the air inlet mode is set to the last user setting.
 - A/C is set to the last user setting.
- For moderate and warm ambient air temperatures as determined by the HVAC module AUTO mode to request heating or cooling:
 - the airflow mode is set to the last user setting.
 - the temperature is set to the last user setting.
 - the blower speed is set to the last user setting.
 - the air inlet mode is set to the last user setting.
 - A/C is set to the last user setting.

System Airflow and Mode Description

AUTO

When AUTO is selected:

- the temperature control setting is manually set to the desired setting.
- the air inlet door is automatically controlled by the HVAC module based on the temperature setting.
- the mode door is automatically controlled by the HVAC module based on the temperature setting.
- the temperature blend doors are automatically controlled by the HVAC module, based on the temperature setting.
- the A/C compressor is automatically controlled by the HVAC module, based on the temperature setting. The A/C compressor cannot operate if the outside temperature is below approximately 0°C (32°F).
- the blower motor is on. The blower motor speed is automatically controlled by the HVAC module based on the temperature setting, but can be manually overridden.

OFF

When OFF is selected:

- the air inlet door closes, preventing outside air and allowing only recirculated air.
- the blower motor is off.

Max A/C

When MAX A/C is selected:

- the air inlet door closes, preventing outside air and allows only recirculated air.
- the RECIRC button is disabled and the indicator is illuminated (RECIRC forced on).
- the mode doors direct airflow to the instrument panel A/C registers.
- blended air temperature is available.
- the A/C request button is illuminated and is disabled.
- the A/C compressor operates if the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

PANEL

When PANEL is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the instrument panel A/C registers.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

PANEL/FLOOR

When PANEL/FLOOR is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the floor duct and the instrument panel A/C registers. A small amount of airflow from the side window demisters and defrost duct is present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

FLOOR

When FLOOR is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment. In this mode, the RECIRC request times-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.
- the mode doors direct airflow to the floor duct. A small amount of airflow from the defroster duct and side window demisters is present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator on) can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor operates and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.

FLOOR/DEFROST

When FLOOR/DEFROST is selected:

- the RECIRC request button is enabled. If the RECIRC request button is selected (indicator on), the air inlet door closes, preventing outside air from entering the passenger compartment. If the RECIRC request button is not selected (indicator off), the air inlet door opens, allowing only outside air into the passenger compartment. In this mode, the RECIRC request times-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.
- the mode doors direct airflow to the floor duct, the defroster duct and the side window demisters.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor operates automatically and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).

- the blower motor is on.

DEFROST

When DEFROST is selected:

- the RECIRC request button and indicator is disabled. Regardless of button and indicator status, the air inlet door opens, allowing only outside air into the passenger compartment.
- the mode doors direct airflow to the defroster duct and side window demisters. A small amount of airflow from the floor duct is present.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor operates automatically and the indicator illuminates if the A/C request button is selected and the outside temperature is above approximately 0°C (32°F).
- the blower motor is on.